VIT | Vidyalankar Institute of Technology

ACCREDITED A+ BY NAAC

LiFi

LIGHTFIDELITY

1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1 0 1 0 0 1 0 1 1 0 0 1 0 1 1
0 1 0 0 1 1 0 1 0 0 1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1 0 1 0 0

Imagine a world where light becomes data.

HOW IT WORKS

LED Bulb
Electrical current → light stream

→

Modulation
High-speed on/off encodes binary

→

LDR Sensor
Detects light pulses → 0s and 1s

→

Data Output
Decoded to text or sound

KEY ADVANTAGES

42
Mbps per LED light
vs. ~7 Mbps shared Wi-Fi

300+
Mbps with 8 lights
for 15 concurrent users

Bi-Di
Full duplex, real-time
upload + download simultaneous

FPGA MINI PROJECT

Transmitter (Edge Artix-7)

ROM
Text / Sound

→

Encoder
ASCII → Binary

↓

LED

Receiver (Edge Artix-7)

LDR Sensor
High=1 / Low=0

→

Decoder
Binary → Output

if text → 7-Seg Display

if audio → Buzzer

~ light ~

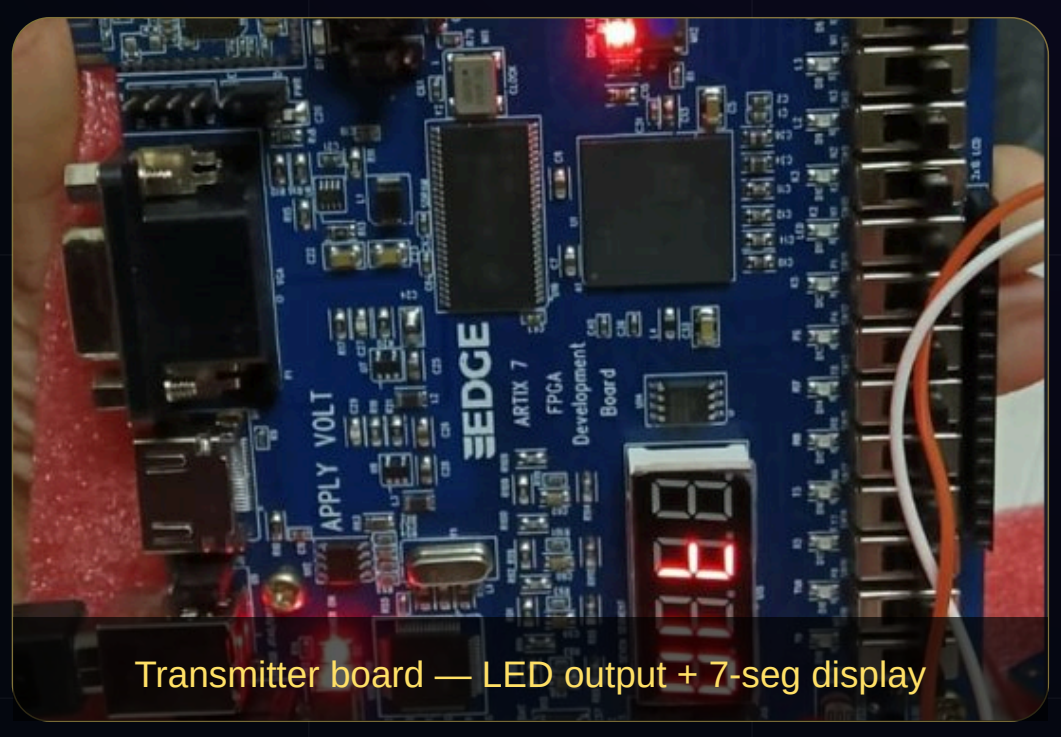
NETWORK FEATURES

- IP per light
- Location tracking
- Geofencing
- Seamless handover
- **Secure**

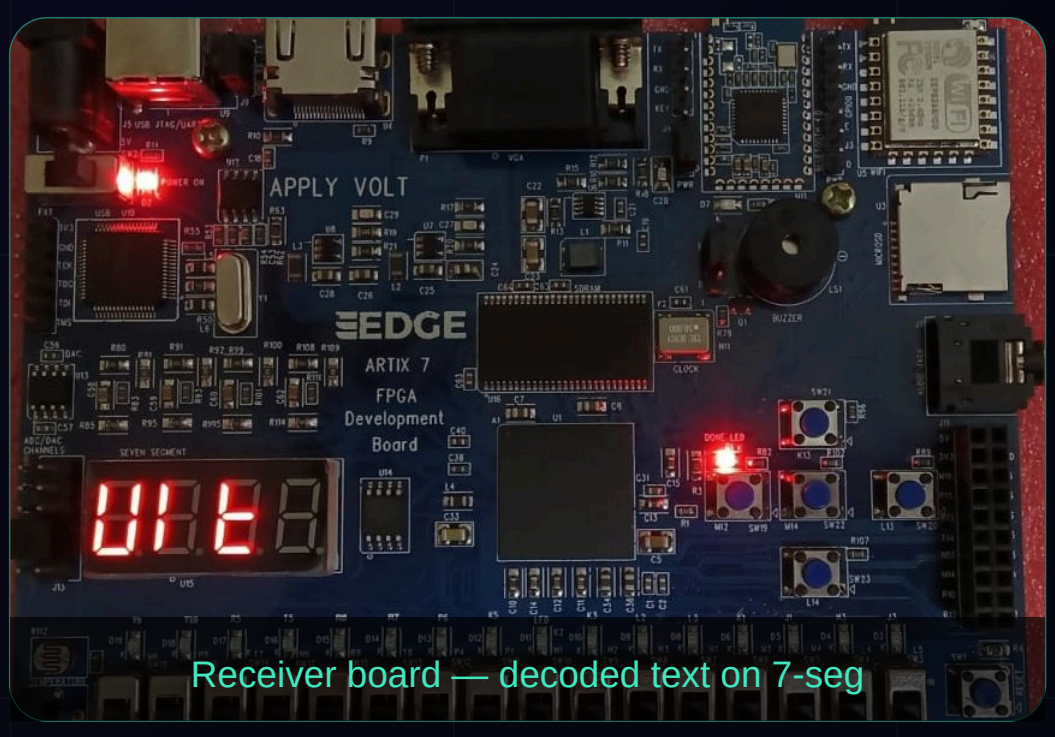
LiFi doesn’t replace your internet provider — it supercharges your last mile.

Once you have a wired connection, deploy LiFi over existing LED lighting using Power over Ethernet or Power Line Communications. Each light = its own node.

HARDWARE IN ACTION — EDGE ARTIX-7 FPGA



Transmitter board — LED output + 7-seg display



Receiver board — decoded text on 7-seg